MID SEMESTER EXAMINATION March'2025

DEPARTMENT OF MATHEMATICS & COMPUTING TECHNOLOGY

Subject: Engg. Maths II (Prob. & Stat) (**MA24101_24102**)

Course : B.Tech (CSE I )+ backlog — *Time:3hrs*

Semester: 2nd sem — *F.M.: 30*

*Answer any Four questions*

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1(a) In an examination 30% of the students failed in physics, 25% in Mathematic and 12% in both Physics and Mathematics. A student is selected at random. Find the probability that

(i) the student has failed in physics, if it is known that that he has failed in Mathematics.

(ii) the student has failed at least one of the two subjects.

(iii) the student has passed at least one of the two subjects.

(iv) the student passed in Mathematics if he failed in Physics.

(b) In a bolt factory machines A,B and C manufacture respectively 25%, 35%, and 40% of the total of their output. 5%,4% and 2% are defective bolts. A bolts is drawn at random from the product and is found to be defective.

What are the probability that it was manufactured by machines A,B and C?

2(a) A random variable $X$ has the following probability mass function,

| $X$ : | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $P(X=x)$: | 0 | $k$ | $2k$ | $2k$ | $3k$ | $k^2$ | $2k^2$ | $7k^2+k$ |

(i) Determine the constant $k$

(ii) Evaluate $P(X<6), P(X\geq 6), P(3<X\leq 6) and\ P3<X/X\leq 6)$

(iii) Find the minimum value of the $x$ ?

(b) Two defective tubes get mixed up with 2 good ones. The tube are tested, one by one, until both defective are found. What is the probability that the last defective tube is obtained on (i) the second test, (ii) the third test and (iii) last the fourth test ?

Find the value of the constant k such that

$$f(x)=\begin{cases} kx(1-x), & 0<x\leq 1 \\ 0, & elswhere \end{cases}$$

Is a possible density function and compute $P\left(X>\frac{1}{2}\right)$ and find $E(x)$

3(a) Find MGF (moment generating function) for X be a discrete random variable having distribution :

| $X$ : | 1 | 2 | 3 | 4 | .... | .... |
|---|---|---|---|---|---|---|
| $f_i$ : | $f_1$ | $f_2$ | $f_3$ | $f_4$ | .... | ..... , |

Where $f_i=\frac{1}{2^i}$

(b) Find MGF (moment generating function) for X be a continuous random variable having distribution given by the p.d.f

$$f(x)=\begin{cases} x, & 0<x\leq 1 \\ 2-x, & 1<x\leq 2 \\ 0, & elswhere \end{cases}$$

4(a) A defective die is thrown ten times independently. The probability that an even number will appear 5 times is twice the probability that an even number will appear 4 times. What is the probability that an even number will appear 4 times. What is the probability that odd face appear in each of the ten throw.

(b) Use Poisson distribution, 2% of the items made by machine are defective. Find the probability that 3 or more items are defective in a sample of 100 items.

(Given $e^{-1}=0.368$ ; $e^{-2}=0.135, e^{-3}=0.498$)

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5(a) The length of bolts produced by a machine is normally distributed with mean 4 and s.d 0.5. A bolt is defective if its length does not lie in the interval (3.8,4.3). Find the percentage of defective bolts produced by the machine.

$$\left\{Given: \ \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0.6} e^{\frac{-t^2}{2}}dt = 0.7257, \quad \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0.4} e^{\frac{-t^2}{2}}dt = 0..6554 \right\}$$

(b) If the skulls are classified as A,B and C according as the length-breadth index is under 75, between 75 and 80, find approximately (assuming that the distribution is normal) the mean and standard deviation of a series in which A are 58%, B are 38% and C are 4%, being given that if

$$\left\{\frac{1}{\sqrt{2\pi}}\int_{0}^{0.2} e^{\frac{-t^2}{2}}dt = 0.08, \ \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{1.75} e^{\frac{-t^2}{2}}dt = 0.46 \right\}$$

8 (a) Find the correlation coefficient between the two variables height (X) and weight(Y) the bivariate data's of 5 person are given by:

| | | | | | |
|---|---|---|---|---|---|
| X: | 64 | 60 | 67 | 69 | 69 |
| Y: | 57 | 60 | 73 | 52 | 68 |

(b) Show by Tchebycheff's inequality that in 2000 throws with a coin the probability that the number of heads lie between 900 and 1100 is at least $\frac{19}{20}$.

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